

Rumeysa Turkmen

PhD Candidate in Computer Science | Concordia University

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PROFESSIONAL SUMMARY

Ph.D. candidate in Computer Science at Concordia University specializing in Virtual Reality, Human-Computer Interaction, and 3D User Interfaces. Authored 11 peer-reviewed publications in top Human-Computer Interaction (HCI) venues (ACM CHI, IEEE ISMAR, IEEE VR, VRST) with 84+ citations and an h-index of 5, plus 4 papers currently under double-blind review. Recipient of the FRQNT doctoral research grant from Fonds de recherche du Quebec and selected as one of 200 young researchers worldwide for the 11th Heidelberg Laureate Forum. Direct-entry PhD admission based on undergraduate research record.

EDUCATION

Doctor of Philosophy (PhD), Computer Science

Sep 2024 – September 2027 (expected)

Concordia University

Montreal, Quebec, Canada

- Direct-entry PhD admission, supervisor: Dr. Anil Ufuk Batmaz
- Research focus: Virtual Reality, Human-Computer Interaction, 3D User Interfaces.
- Awarded FRQNT doctoral research grant (Fonds de recherche du Quebec – Nature et technologies).

Bachelor's Degree, Electrical and Electronics Engineering

2020 – Jun 2024

Kadir Has University

Istanbul, Turkiye

- GPA: 3.47 / 4.00 — Graduated with Honor Degree.
- Full scholarship recipient throughout undergraduate studies.
- Graduation project received “Second Best Design Project” award.

HONORS & AWARDS

- FRQNT Doctoral Research Grant — Fonds de recherche du Quebec – Nature et technologies (2026).
- 11th Heidelberg Laureate Forum — Selected as one of 200 young researchers worldwide (2023).
- Study in Canada Scholarship — Government of Canada (funded research internship at Dalhousie University) (2022).
- Full Undergraduate Scholarship — Kadir Has University (4 years) (2020-2024).
- “Second Best Design Project” award — Kadir Has University graduation project (2024).
- Honor Degree graduate — Kadir Has University.

RESEARCH & TEACHING EXPERIENCE

Teaching Assistant

Sep 2024 – Present

Concordia University

Montreal, Quebec, Canada

- COMP 376 — Introduction to Game Development.
- COMP 476 — Advanced Game Development.

Research Assistant — EXIT Lab

Sep 2024 – Present

Concordia University

Montreal, Quebec, Canada

- Conducting doctoral research on controller-free interaction such as eye-gaze and voice for 3D selection and manipulation techniques in Virtual and Augmented Reality.
- First-author publications at top HCI venues.

Research Intern

Jul 2023 – Jun 2024

TUBITAK (Scientific and Technological Research Council of Türkiye)

Türkiye

- Led research on the TUBITAK-1002 funded project: "Input and Evaluation Methods of Eye-Gaze Behaviors to be Used on Virtual Reality Applications," supervised by Dr. Mine Sarac Stroppa.

Visiting Research Student (Undergraduate)

Sep 2022 – Jan 2023

Dalhousie University — VERTEX Lab (HCI)

Halifax, Nova Scotia, Canada

- Conducted HCI research under Dr. Mayra Barrera Machuca's supervision, funded by the Study in Canada Scholarship.
- Co-authored work on 3D pointing and 3D sketching in VR.

Undergraduate Research Assistant — HapLab

Sep 2021 – Jun 2024

Kadir Has University

Istanbul, Türkiye

- Researched eye-gaze behavior in 3D systems and haptic interaction.
- Foundation for multiple first- and co-authored publications in VR/HCI venues.

PUBLICATIONS

84+ citations, h-index 5, i10-index 4 (Google Scholar, 2026).

Peer-Reviewed Conference Publications

1. **R. Turkmen**, Z. Rasoulifar, A. U. Batmaz. "VoiceRay: 3D Object Selection Technique for Occluded and Dense Environments in Virtual Reality." ACM CHI Conference on Human Factors in Computing Systems, 2026.
2. **R. Turkmen**, L. Voisard, M. Kersten-Oertel, A. U. Batmaz. "Adaptive Hand Visibility for Accurate 3D User Interactions in Virtual Environments." IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2025.
3. **R. Turkmen**, F. R. Ortega, A. U. Batmaz. "Don't Miss Notifications: Exploring Gaze Notifications for Virtual Reality Cooking Environment." IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2025.
4. **R. Turkmen**, Z. E. Gelmez, A. U. Batmaz, W. Stuerzlinger, P. Asente, M. Sarac. "Eyeguide & Eyeconguide: Gaze-based Visual Guides to Improve 3D Sketching Systems." Proceedings of the 2024 ACM CHI Conference on Human Factors in Computing Systems, 2024.
5. A. U. Batmaz, **R. Turkmen**, M. Sarac, M. D. Barrera Machuca, W. Stuerzlinger. "Re-investigating the Effect of the Vergence-Accommodation Conflict on 3D Pointing." Proceedings of the 29th ACM Symposium on Virtual Reality Software and Technology (VRST), 2023.
6. A. U. Batmaz, **R. Turkmen**, M. Sarac, M. D. B. Machuca, W. Stuerzlinger. "Effect of Grip Style on Peripersonal Target Pointing in VR Head Mounted Displays." IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2023.
7. **R. Turkmen**, K. Pfeuffer, M. D. Barrera Machuca, A. U. Batmaz, H. Gellersen. "Exploring Discrete Drawing Guides to Assist Users in Accurate Mid-Air Sketching in VR." CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI EA), 2022.
8. **R. Turkmen**, A. U. Batmaz. "Cezve: A Virtual Turkish Coffee Pot Interaction Technique for 3D Target Selection in Virtual Reality." IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR) Abstracts and Workshops, 2026.

9. **R. Turkmen**, N. Hatira, R. J. Teather, M. Kersten-Oertel, W. Stuerzlinger, et al.. *"Investigating Adaptive Hand Visibilities for Accurate 3D User Interactions in Augmented Reality."* IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR) Abstracts and Workshops, 2026.
10. P. Rawat, **R. Turkmen**, C. Nwagu, K. Sunday, M. D. B. Machuca. *"Evaluating Voxel-Based Graphical Passwords for Virtual Reality."* IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR) Abstracts and Workshops, 2024.
11. **R. Turkmen**, C. Nwagu, P. Rawat, P. Riddle, K. Sunday, M. B. Machuca. *"Put Your Glasses On: A Voxel-based 3D Authentication System in VR Using Eye-Gaze."* IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR) Abstracts and Workshops, 2023.

Under Review (Double-Blind)

4 papers (2 as first-author, 2 as second-author) currently under double-blind peer review at top HCI/VR venues. Details available on request.

RESEARCH AREAS & SKILLS

Research Areas: Virtual Reality, Augmented Reality, Human-Computer Interaction, 3D User Interfaces, Eye-Gaze Interaction, Voice interaction, 3D Selection & Pointing, Haptic Interaction, Game Development.

Technical: Experience for 5+ years on Unity, C#, VR/AR development, eye-tracking systems, user study design, quantitative & qualitative analysis, statistical methods and analyses.

Languages: English (fluent), Turkish (native), French (A2).

SERVICE & VOLUNTEERING

- Student Volunteer — IEEE ISMAR 2023, 2025, ACM CHI 2022 and 2024.
- Climate Representer — Ministry of Environment and Urbanization, Türkiye, Jan 2022 – Sep 2022.
- President — Environment and Zero Waste Club, Mar 2021 – Jun 2022.